

_____ 1) Does music affect your pulse? A study was done to determine if the pulses of the subjects were significantly higher after listening to pop music versus listening to classical piano music. The subjects were assigned to one of two treatments: 30 minutes of pop music or 30 minutes of classical piano music. The pulse of each subject was recorded after listening to the music. Let μ_p represent the mean pulse after listening to pop music and μ_c represent the mean pulse after listening to classical piano music. Which of the following sets of hypotheses would be appropriate to determine if the mean pulse after listening to pop music is significantly higher than the mean pulse after listening to classical piano music?

- (A) $H_0: \mu_p > \mu_c$ vs $H_a: \mu_p = \mu_c$
- (B) $H_0: \mu_p = \mu_c$ vs $H_a: \mu_p < \mu_c$
- (C) $H_0: \mu_p = \mu_c$ vs $H_a: \mu_p > \mu_c$
- (D) $H_0: \mu_p < \mu_c$ vs $H_a: \mu_p > \mu_c$
- (E) $H_0: \mu_p = \mu_c$ vs $H_a: \mu_p \neq \mu_c$

Use the following situation for Questions 2 – 4:

A study of iron deficiency among infants compared blood hemoglobin levels of a random sample of one year old infants who had been breast-fed to a random sample of one year old infants who had been fed with standard infant formula. Here are the results

	n	$\bar{x}$	s
Breast-fed infants	23	13.3	1.7
Formula-fed infants	19	12.4	1.8

We wish to test the hypotheses: $H_0: \mu_B = \mu_F$ and $H_A: \mu_B \neq \mu_F$ where μ_B and μ_F are the population mean blood hemoglobin levels for breast-fed and formula-fed infants, respectively.

_____ 2) What additional information would you need to confirm that the conditions for this test have been met?

- (A) Each sample distribution of infant groups has no outliers or strong skewness.
- (B) Each population distribution of infant groups follows a normal population.
- (C) The combined sample distribution of infant groups has no outliers or strong skewness.
- (D) Both A and B are true.
- (E) Both B and C are true.

_____ 3) Which of the following represents the correct t-test statistic and the p-value for this inference procedure?

- (A) $t = 1.65$ and $p\text{-value} = 0.0581$
- (B) $t = 1.65$ and $p\text{-value} = 0.1162$
- (C) $t = 2.19$ and $p\text{-value} = 0.0210$
- (D) $t = 2.19$ and $p\text{-value} = 0.0419$
- (E) None of the above give a complete answer

_____ 4) Which of the following correctly interprets this p-value in the context of the study?

- (A) The p-value represents the probability we get a difference in blood hemoglobin levels this large or larger if there is no difference between breastfed babies and formula fed babies.
- (B) The p-value represents the probability we get a difference in blood hemoglobin levels this small or smaller if there is not difference between breastfed babies and formula fed babies.
- (C) The p-value represents the probability we get this difference in blood hemoglobin levels or more extreme if there is no difference between breastfed babies and formula fed babies.
- (D) The p-value represents the probability that there is no difference between breastfed babies and formula fed babies hemoglobin levels if we get a difference like this or larger.
- (E) The p-value represents the probability that there is no difference between breastfed babies and formula fed babies hemoglobin levels if we get a difference like this or smaller.

_____ 5) A company wants to compare the average productivity of employees working from home versus those working in the office. They measure productivity as the number of tasks completed per week. They randomly select 12 employees working from home and 12 employees working in the office and record their number of weekly task completions. Which of the following is NOT a condition that should be checked before running a significance test for a mean difference in productivity levels?

- (A) Each of the samples are approximately Normally distributed.
- (B) The home sample is independent of the working in office sample.
- (C) The people in each sample were selected at random.
- (D) No more than 10% of each population was sampled.
- (E) A graph of each sample data has no outliers or strong skewness.